

Pretrained AI/ML Models and Resources for Smart Agriculture DSS

Technical integration reference guide for engineering final-year project development. Designed for direct model acquisition, API orchestration, and deployment.

Target: Engineering DSS Implementation **Deployment:** Edge / Cloud Hybrid **Language Focus:** Kannada & English

1. Technical Classification Framework

Before downloading assets, distinguish between the distinct resource types to prevent architectural errors:

1. Ready Pretrained

Weights trained on target tasks. Load directly via PyTorch/ONNX for instant inference without retraining.

2. Download & Fine-Tune

Base backbones (e.g., YOLO, MobileNet) initialized on ImageNet/COCO; transfer learning needed on domain data.

3. Training Dataset

Raw tabular or image data. Requires building an ML pipeline from scratch (e.g., Random Forest, XGBoost).

4. Managed API

Cloud-hosted endpoints (e.g., Open-Meteo, Bhashini). Returns live data; weights cannot be saved locally.

2. Module Specifications & Recommended Models (Modules 1 – 3)

Engineering Notice (Module 1): Universal pretrained models for crop recommendation do NOT exist reliably. Soil-crop interactions vary drastically across agro-ecological zones. Train a localized tabular classifier using verifiable agronomic metrics.

Module	Model / Asset	Function & Inputs	Source	Direct Link	Cost	License	Pipeline & Recommendation
1. Crop Recommendation	Random Forest / XGBoost (Custom Trained)	Predicts optimal crop for field conditions. Inputs: N, P, K (kg/ha), soil pH, temp (°C), humidity (%), rainfall (mm).	Kaggle / Agronomy Data	Kaggle Dataset	Free	Open Data (CDLA-Permissive)	CSV → pandas → Scikit-Learn Random Forest Classifier → pickle.dump() → FastAPI. Rank 1
2. Crop Disease Detection	MobileNetV3-Small / InceptionResNetV2	Classifies foliar pathology across 38 crop-disease classes from RGB leaf photos (224×224).	Hugging Face / Keras	HF Repo / PlantDoc	Free	Apache 2.0 / CC BY 4.0	Leaf Photo → Resize 224×224 → MobileNetV3 TFLite inference → Disease Label + Confidence. Rank 1 (Mobile)
3. Pest Detection	YOLOv8 Nano (yolov8n.pt) (Fine-Tuned)	Localizes and bounds insect pests (aphids, stem borers, whiteflies) in field foliage images.	Ultralytics / GitHub	Ultralytics Repo	Free	AGPL-3.0 (Open Source)	Image → YOLOv8 fine-tuned on IP102 pest dataset → Bounding boxes + Class counts. Rank 1

*Note: PlantVillage provides 54k laboratory images. Fine-tune on the in-field *PlantDoc* dataset for robust real-world field accuracy.

2. Module Specifications & Recommended Models (Modules 4 – 8)

Engineering Notice (Modules 5 & 6): Irrigation and fertilizer scheduling must NOT rely on black-box neural networks. Agricultural DSS best practice combines ML predictions with FAO-56 Penman-Monteith water equations and ICAR/STCR fertilizer rule engines.

Module	Model / Asset	Function & Inputs	Source	Direct Link	Cost	License	Pipeline & Recommendation
4. Yield Prediction	XGBoost / LightGBM Regressor	Estimates crop yield (Quintals/Acre) from historical rain, GDD (Growing Degree Days), fertilizer, and acreage.	Kaggle / Data.gov.in	Data.gov.in / XGBoost Repo	Free	Apache 2.0 / Open Govt	Train on district crop yield history → Save XGBoost JSON → Serve via Python API. Rank 1
5. Smart Irrigation	Hybrid: FAO-56 Engine + Random Forest	Calculates Evapotranspiration (ET_0) and crop coefficient (K_c) to output required irrigation volume (liters/plot).	Open-Meteo + PyETo	Open-Meteo / PyETo Library	Free	CC-BY 4.0 / MIT	Weather API → Compute $ET_c = ET_0 \times K_c$ → Soil moisture balance → Valve trigger duration. Rank 1
6. Fertilizer Advisory	STCR Rule Engine + Decision Tree	Calculates precise N-P-K fertilizer bags needed based on Soil Test Crop Response (STCR) targeted yield equations.	ICAR Guidelines / Scikit-Learn	ICAR Portal / Scikit-Learn	Free	Public Domain / BSD	Soil test values + Target yield → Apply agronomic target formula → Output commercial urea/DAP/MOP bags. Rank 1
7. Weather & Risk Forecasting	Open-Meteo API + GBDT Frost/Rain Classifier	Provides 7-day hourly forecasts; GBDT model flags localized pest outbreak, heat stress, or excessive rain risk.	Open-Meteo / ECMWF Data	Open-Meteo Docs	Free (Non-comm)	CC BY 4.0	GPS coordinates → Fetch JSON forecast → Run risk rules (e.g. humidity > 90% for 48h = fungal alert). Rank 1
8. Market Price Forecasting	NeuralProphet / Auto-ARIMA	Forecasts 15–30 day commodity modal mandi prices (₹/Quintal) capturing market seasonality and arrival trends.	Agmarknet / PyPI	Agmarknet / NeuralProphet	Free	Apache 2.0 / Govt Data	Scrape/download APMC historical prices → Fit time series model → Output expected trend & confidence interval. Rank 1

Algorithmic Decision Support Integration Flow

[Live Weather API (Open-Meteo)] + [Soil Sensor/Test (NPK, pH)] + [Crop Phenology Stage]

↓

Decision Support Engine: Agronomic Balance (FAO-56 ET_0 + STCR Target Equation)

↓

Output: "Irrigate 18mm today; Apply 25kg Urea at 45 DAS; Fungal blight risk: HIGH"

2. Module Specifications & Recommended Models (Modules 9 – 12)

Language & Remote Sensing Strategy: Multilingual support must prioritize Kannada and Indian English. For satellite observation, use Sentinel-2 open optical bands processed via NDVI algorithms rather than resource-heavy generic vision transformers.

Module	Model / Asset	Function & Inputs	Source	Direct Link	Cost	License	Pipeline & Recommendation
9. Agriculture Chatbot	Llama-3-8B-Instruct (RAG Grounded) / Qwen2.5-7B	Answers farmer queries in Kannada and English grounded on agricultural university extension packages of practices.	Meta / Hugging Face	Llama-3-8B / Qwen2.5-7B	Free Open-Weight	Llama 3 Community / Apache 2.0	User Query → FAISS vector search on Agri PDFs → Ingest prompt + context into 4-bit Ollama/vLLM backend. Rank 1
10. Voice Input (STT)	IndicWav2Vec / OpenAI Whisper-Small	Converts spoken Kannada & Indian English farmer audio into transcribed UTF-8 text strings.	AI4Bharat / Hugging Face	AI4Bharat / Whisper-Small	Free	MIT License	App captures .wav audio (16kHz) → Whisper-Small PyTorch endpoint → Transcribed text passed to chatbot. Rank 1
10b. Voice Output (TTS)	Indic-Parler-TTS / Edge-TTS	Synthesizes natural-sounding Kannada and English audio advice from DSS text recommendations.	AI4Bharat / Microsoft	Indic-Parler / Edge-TTS	Free	MIT / Apache 2.0	DSS text result → Edge-TTS Python service → Return MP3 stream to mobile app speaker. Rank 1
11. Satellite / Remote Sensing	Sentinel-2 NDVI Pipeline / Prithvi-EO-2.0	Computes crop vigor, NDVI, and water stress index from 10m multispectral satellite bands (B4-Red, B8-NIR).	Copernicus / IBM-NASA HF	Prithvi-EO / GEE Python	Free Academic	Apache 2.0 / Open Science	Polygon bounds → Fetch Sentinel-2 GeoTIFF → Rasterio calculates (NIR-Red)/(NIR+Red) → Field health map. Rank 1
12. IoT & Sensor Layer	Kalman Filter + Threshold Heuristics	Filters noisy soil NPK, moisture, and temperature readings from ESP32 field nodes via MQTT.	Eclipse Mosquitto / Scipy	Mosquitto / SciPy Signal	Free	EPL / BSD	ESP32 → MQTT broker → Python listener → 1D Kalman smoothing → Commit clean telemetry to PostgreSQL. Rank 1

3. Complete Smart Agriculture DSS Architecture

The system is designed as a distributed modular decision engine where raw ML model outputs are contextualized by domain agronomy logic before reaching the farmer.

Farmer Client (Flutter / Mobile UI - Kannada & English)
 ↓ (REST / WebSocket Audio & Payloads)
 Backend API Gateway (FastAPI Python / Rust) (Auth & Validation)

4. Detailed Model Integration Blueprints

Step-by-step developer specifications for the four primary models our team will implement into the FastAPI microservice backend.

Crop Recommendation (Scikit-Learn Random Forest)

Format: crop_rf_model.pkl / Joblib
Inputs: Float Array [N, P, K, Temp, Humidity, pH, Rainfall]
Output: Predicted Crop Class (e.g., "Ragi", "Maize", "Cotton")
Hardware: Standard CPU laptop (< 50MB RAM, < 2ms latency)
Mobile Viable: Yes (Can run via ONNX Runtime inside Flutter)
Fine-Tuning: Required: Fit model on regional Karnataka soil records.
Code Pipeline:

```
clf = joblib.load('crop_rf.pkl')
pred = clf.predict_proba([[90, 42, 43, 24.5, 82.1, 6.5, 202.8]])
```

Crop Disease Detection (MobileNetV3 TFLite)

Format: plant_disease_mobilenet.tflite (Quantized INT8)
Inputs: Leaf Camera Image [1, 224, 224, 3] Float32 / 255.0
Output: Softmax vector of 38 disease classes + confidence %
Hardware: Runs on standard CPU laptop or on Android directly
Mobile Viable: Yes (Direct on-device inference without internet)
Fine-Tuning: Recommended on PlantDoc dataset to reduce lab bias.
Code Pipeline:

```
interpreter = tflite.Interpreter('model.tflite')
interpreter.set_tensor(in_idx, img); res = interpreter.get_tensor(out_idx)
```

Pest Object Detection (YOLOv8 Nano)

Format: yolov8n_pest.pt (Convertible to .onnx / .tflite)
Inputs: Camera RGB Frame [640, 640, 3]
Output: Bounding boxes [x1, y1, x2, y2], class ID, confidence score
Hardware: CPU laptop runs at ~15-20 FPS; GPU runs > 90 FPS
Mobile Viable: Yes via Flutter TFLite plugin or ONNX Runtime
Fine-Tuning: Mandatory on pest dataset (COCO pretrained detects general objects).
Code Pipeline:

```
model = YOLO('yolov8n_pest.pt')
results = model.predict(source='leaf.jpg', conf=0.45)
```

Multilingual Agri Chatbot (Llama-3-8B RAG)

Format: GGUF 4-bit Quantized (llama-3-8b-instruct.Q4_K_M.gguf)
Inputs: Farmer Prompt + Retrieved ICAR Extension Manual Snippets
Output: Clean, verified agricultural guidance in Kannada/English
Hardware: 8GB VRAM GPU or 16GB RAM laptop via Ollama / llama.cpp
Mobile Viable: No (Server-side API execution required)
Fine-Tuning: Use RAG (Retrieval-Augmented Generation) instead of fine-tuning.
Code Pipeline:

```
ollama.chat(model='llama3', messages=[{'role': 'user', 'content': prompt}])
```

5. Recommended Free Tech Stack for Final-Year Project

Layer / Function	Recommended Tool / Technology	Why It Fits Our Student Project
Model Sources & Training	Hugging Face Hub + Kaggle Kernels	Free T4/P100 GPU quota (30 hrs/week on Kaggle) for training models without paying cloud costs.
ML Frameworks	PyTorch, Ultralytics, Scikit-Learn	Standard open-source Python libraries with abundant community examples and easy export options.
Backend API Service	FastAPI (Python 3.10+) + Uvicorn	High throughput asynchronous endpoints, auto Swagger UI docs, and native typing for data payloads.
Mobile & Web UI	Flutter (Dart) / React Native	Single codebase runs natively on Android smartphones and web browsers; excellent camera plugins.
Relational Database	PostgreSQL / Supabase (Free Tier)	Stores farmer profiles, crop records, mandi time series, and field history reliably with zero licensing fee.

6. Resource Economics: Free vs Paid Infrastructure

Pretrained model weights are universally free to download. Financial expenses only occur when utilizing commercial managed cloud hardware. For our student project, zero capital expenditure is required.

Platform / Asset	Free Tier Scope	What the Free Version Provides	When Payment Triggers	Student Project Need
Hugging Face Hub	100% Free Public	Unlimited download of open models, datasets, and local checkpoints.	Private repos, dedicated Inference Endpoints (\$0.60/hr).	NOT REQUIRED
Kaggle / Colab	Free with Limits	Weekly GPU hours (T4/P100) suitable for fine-tuning YOLO and MobileNet.	Colab Pro+ subscriptions, continuous non-preemptible jobs.	NOT REQUIRED
Open-Meteo Weather	Free Non-Commercial	Up to 10,000 daily API calls, hourly 7-day forecast, ERA5 history.	Commercial redistributions, >10,000 requests per day.	NOT REQUIRED
OpenAI / Claude API	Paid Pay-As-You-Go	High reasoning LLMs (GPT-4o), but incurs direct dollar billing.	Every single prompt token and response token.	AVOID (Use Llama 3)
Bhashini Speech APIs	Free Academic Access	Government of India national language translation, STT, and TTS.	High enterprise concurrency limits.	NOT REQUIRED

7. Quality Verification: Checklist Before Model Download

Adhere strictly to these evaluation rules before committing any external checkpoint to our code repository:

- 1. Verify Model License:** Ensure the model is released under Apache 2.0, MIT, CC-BY, or academic open-access.
- 2. Scrutinize Accuracy Claims:** Treat claims like "99.8% plant disease accuracy" with caution. High laboratory accuracy on uniform backgrounds often collapses to < 60% on complex field leaf backgrounds.
- 3. Evaluate Hardware Footprint:** Check parameter counts. Avoid models > 1GB if a 20MB MobileNet or YOLOv8n delivers sufficient precision for field deployment.
- 4. Validate on Local Data:** Run test inferences against crop diseases and soil types common to Karnataka before incorporating recommendations into user-facing advice.

8. Teammate Implementation Action Checklist

☐ Step 1: Download Crop Recommendation CSV from Kaggle.	☐ Step 2: Train & serialize Scikit-Learn Random Forest model.
☐ Step 3: Download MobileNetV3 / InceptionResNet plant model.	☐ Step 4: Fine-tune weights on PlantDoc in-field dataset.
☐ Step 5: Quantize disease classifier to TFLite (INT8 format).	☐ Step 6: Download YOLOv8n and fine-tune on pest images.
☐ Step 7: Test Open-Meteo REST API calls with local GPS.	☐ Step 8: Implement FAO-56 Penman-Monteith water logic.
☐ Step 9: Pull historical Agmarknet modal commodity prices.	☐ Step 10: Setup Ollama with Llama-3-8B-Instruct locally.
☐ Step 11: Connect Edge-TTS / Indic-TTS for Kannada audio output.	☐ Step 12: Package models inside FastAPI microservice endpoints.